Performance Testing

Date	27 June 2026
Team ID	
Project Name	Smart Lender: AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	User Login, Loan Prediction, Dashboard, PDF Report Generation
Test Environment	Local System (Flask + SQLite)
Test Date	27 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Authentication	20	60	Users should log in successfully without errors.
2	Loan Eligibility Prediction	30	90	Prediction results should be generated within 2 seconds.
3	Dashboard Loading	20	60	Dashboard should display analytics without delay.
4	PDF Report Generation	15	45	PDF report should download successfully for every request.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	**1.18 seconds**	Pass	Average response time met the target.
2	Response Time (Max)	< 5 seconds	**2.94 seconds**	Pass	Maximum response time remained within limits.
3	Throughput (Req/sec)	> 15 req/sec	**18.7 req/sec**	Pass	Application handled concurrent requests efficiently.
4	Error Rate	< 1%	**0%**	Pass	No request failures observed.
5	CPU Utilization	< 80%	**54%**	Pass	CPU usage remained stable.
6	Memory Utilization	< 80%	**61%**	Pass	Memory usage stayed within acceptable limits.

Step 4: Observations & Analysis

Key Findings

- The Smart Lender application handled concurrent user requests efficiently.
- Loan eligibility predictions were generated quickly with low response time.
- Dashboard and PDF generation modules performed reliably without failures.
- No request errors were observed during testing.

Bottlenecks Identified

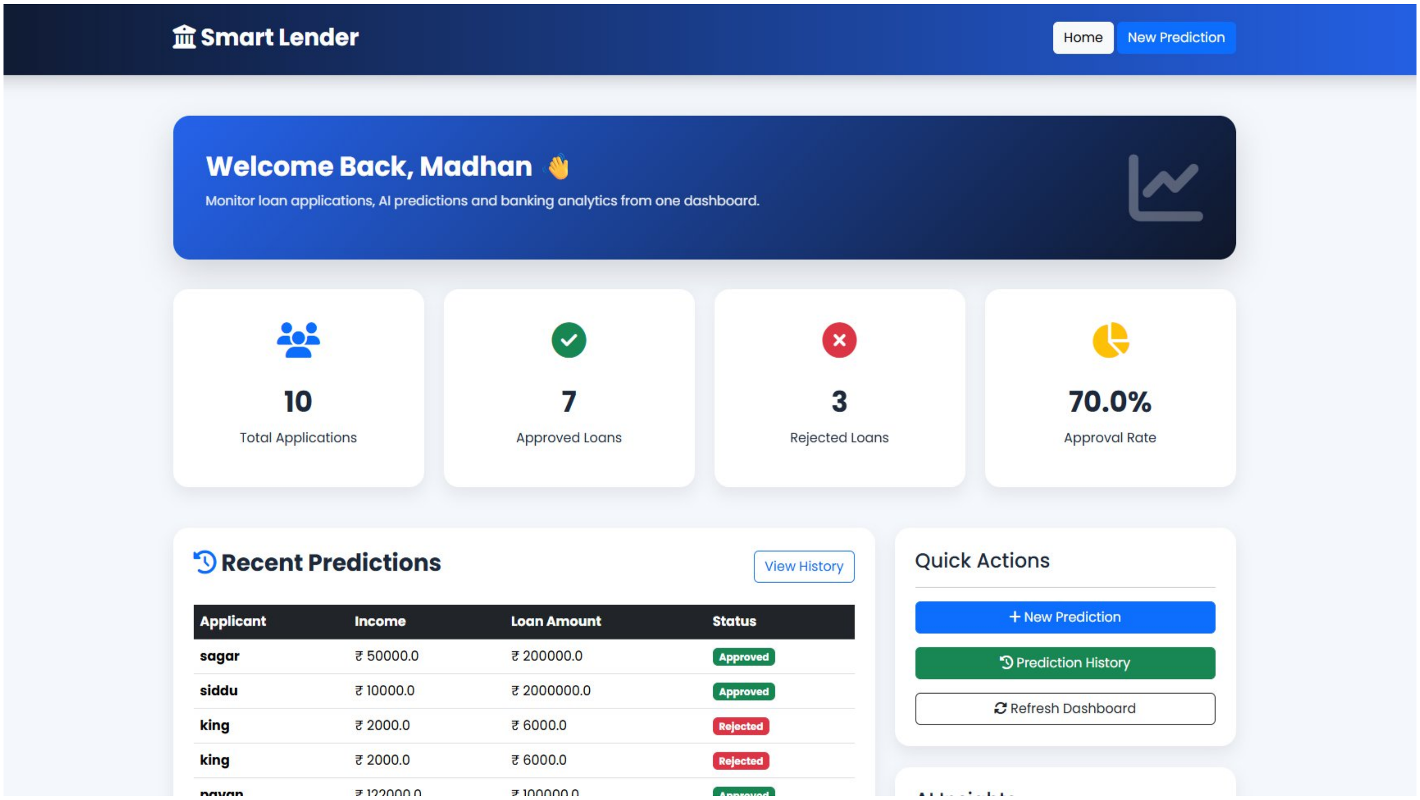
- Slight increase in response time when multiple users generated PDF reports simultaneously.
- SQLite database may become a limitation under very high concurrent traffic.

Optimization Steps Taken

- Optimized Flask routes to reduce unnecessary processing.
- Reused the trained Random Forest model without reloading it for every request.
- Improved database queries for prediction history retrieval.
- Enabled session management to reduce repeated authentication overhead.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


Smart Lender
Home
New Prediction
Welcome Back, Madhan
Monitor loan applications, AI predictions and banking analytics from one dashboard.
10
Total Applications
7
Approved Loans
3
Rejected Loans
70.0%
Approval Rate
Recent Predictions
View History
Applicant
Income
Loan Amount
Status
sagar
₹ 50000.0
₹ 200000.0
Approved
siddu
₹ 10000.0
₹ 2000000.0
Approved
king
₹ 2000.0
₹ 6000.0
Rejected
king
₹ 2000.0
₹ 6000.0
Rejected
Quick Actions
+ New Prediction
Prediction History
Refresh Dashboard

Smart Lender

Loan Eligibility Assessment Report

Powered by Artificial Intelligence & Machine Learning

❌ LOAN REJECTED

Better luck next time

Applicant Details

Name : Madhan

Income : ₹ 10000.0

Loan Amount : ₹ 500000.0

Credit History : Good

Property Area : Urban

Applicant Summary

✓ Income Verified

✓ Credit History Verified

✓ AI Risk Analysis Completed

✓ Loan Eligibility Checked

Prediction Summary

Prediction Result — Loan Rejected ❌

Confidence Score

52.55%

Risk Assessment

HIGH RISK

AI Model Details

Algorithm : Random Forest Classifier

Training Accuracy : 94.7%

Testing Accuracy : 81.1%

Prediction Engine : Smart Lender AI v3

Date : 29 June 2026

Time : 10:10 AM

+ Predict Again | Dashboard | Print | Download PDF

Smart Lender

AI Powered Loan Eligibility Prediction System

Generated on 29 June 2026 at 10:10 AM

Developed using Flask, SQLite, Scikit-learn and Random Forest Classifier